PHY 130A
Lecture Notes:
Fermi's Golden Rule

Michael Mulhearn

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1 The Interaction Picture

We consider a time-dependent Hamiltonian of the form:

$$H = H^{(0)} + \delta H(t)$$

where $H^{(0)}$ is time independent and so well understood, with energy eigenstates:

$$H^{(0)} |n\rangle = E_n |n\rangle$$

and the $\delta H(t)$ represents a new interaction. The new interaction is time dependent and therefore has no energy eigenstates.

The interaction picture factors out the simple and well-understood time dependence from $H^{(0)}$ from the to-be-determined time dependence of $\delta H(t)$ as:

$$|\Psi(t)\rangle = \sum_m c_m(t) \exp\left(-i\frac{E_m t}{\hbar}\right)|m\rangle$$

The $\{|m\rangle\}$ are still a basis for the full Hamiltonian H, even though they are not eigenstates, so this is certainly a valid expansion, due to the time dependent coefficients $c_m(t)$. This just says that $|\Psi(t)\rangle$ can be expressed at any time as a sum of basis vectors, which is certainly true. We will know it is useful if it simplifies the problem. Specifically, if the time dependence in the coefficients is due exclusively to $\delta H(t)$.

Applying the SE:

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle &= H\,|\psi(t)\rangle \\
&= H^{(0)}\,|\psi(t)\rangle + \delta H(t)\,|\psi(t)\rangle \\
\left(i\hbar\frac{\partial}{\partial t} - H^{(0)}\right)|\psi(t)\rangle &= \delta H(t)\,|\psi(t)\rangle
\end{aligned}$$

where we have arranged things in the hope of getting some cancellation on the left hand side:

$$\begin{aligned}
\left(i\hbar\frac{\partial}{\partial t} - H^{(0)}\right)|\psi(t)\rangle &= \sum_m \{i\hbar\dot{c}_m(t) + E_m c_m(t) - E_m c_m(t)\}\exp\left(-i\frac{E_m t}{\hbar}\right)|m\rangle \\
&= \sum_m i\hbar\dot{c}_m(t)\,\exp\left(-i\frac{E_m t}{\hbar}\right)|m\rangle
\end{aligned}$$

and indeed we do. Meanwhile the RHS is:

$$\delta H(t)\,|\psi(t)\rangle = \sum_m c_m(t)\,\exp\left(-i\frac{E_m t}{\hbar}\right)(\delta H(t)\,|m\rangle)$$

and so we cannot match coefficients term by term in the expansion because:

$$|m\rangle \;\neq\; \delta H(t)\;|m\rangle$$

So changing dummy variables to prepare for what follows:

$$\delta H(t)\,|\psi(t)\rangle = \sum_n c_n(t)\,\exp\left(-i\frac{E_n t}{\hbar}\right)(\delta H(t)\,|n\rangle)$$

And inserting a complete set:

$$\begin{aligned}(\delta H(t)\;|n\rangle) &= \left(\sum_m |m\rangle\langle m|\right)(\delta H(t)\,|n\rangle)\\ &= \sum_m \langle m|\delta H(t)|n\rangle\,|m\rangle\end{aligned}$$

And now putting back LHS equal to RHS

$$\sum_m i\hbar\dot{c_m}(t)\,\exp\left(-i\frac{E_m t}{\hbar}\right)|m\rangle = \sum_m\sum_n c_n(t)\,\exp\left(-i\frac{E_n t}{\hbar}\right)\langle m|\delta H(t)|n\rangle$$

Where we can now match term by term to obtain:

$$\dot{c_m}(t) = -\frac{i}{\hbar}\sum_n c_n(t)\,\exp\left(i\frac{(E_m - E_n)t}{\hbar}\right)\langle m|\delta H(t)|n\rangle\,|m\rangle \tag{1}$$

This is everything we hoped for! Because of the time depedence of the total Hamiltonian, our expansion coefficients have to be time dependent as well (the vectors $|n\rangle$ are still a basis, but they are not stationary states of the total Hamiltonian H). But, in the interaction picture [1] we factor out the time dependence due to $H^{(0)}$, and do not include it in $c_m(t)$. Therefore, the time dependence of the $c_m(t)$ does not depend on $H^{(0)}$.

2 First-Order Time-Dependent Perturbation Theory

For our purposes, it suffices to consider the relatively simple time-dependent interaction $\delta H(t)$ shown in Fig. 1. Here the interaction potential V has no explicit time dependence, but it is turned on at t=0 and then off at $t = T$. We will also assume that the initial state is an energy eigenstate $|i\rangle$ with

$$H^{(0)}\,|i\rangle = E_i\,|i\rangle$$

[1]Formally, in the interaction picture the operators evolve as

$$O(t) = \exp(iH^{(0)}t)\;O\;\exp(-iH^{(0)}t)$$

while the states carry the remaining time dependence. Our derivation here achieves exactly the same separation of time dependence without invoking time evolution of the operators.

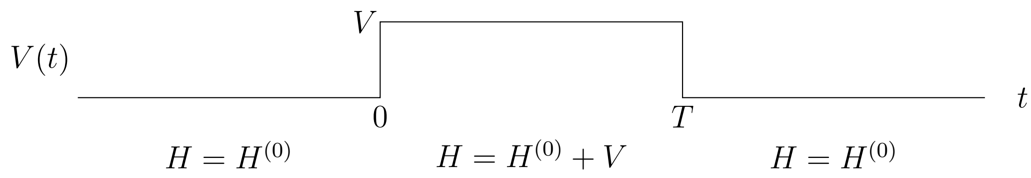

Figure 1: The interaction potential V is time-independent, however, it turns on at $t = 0$ and then off at $t = T$, making the total Hamiltonian time-dependent.

Before $t \leq 0$, this is a stationary state of total Hamiltonian, because the interaction potential is off, and we have the familiear time dependence:

$$|\Psi(t)\rangle = \exp\left(-i\frac{E_m t}{\hbar}\right)\,|i\rangle \qquad\qquad (t \leq 0)$$

This implies that:

$$c_m(0) = \delta_{mi}$$

However, starting at $t = 0$, the interaction turns on and $|i\rangle$ is no longer a stationary state. The probability p of observing the state $|m\rangle$ is given by:

$$p(t) = |c_m(t)|^2 \qquad\qquad (0 \leq t \leq T)$$

which we can no longer assume is zero for $m \neq i$, and is time dependent. At $t = T$, the interaction turns off. The states $|m\rangle$ are again Eigenstates, but the state is no longer $|i\rangle$. We now have a probability

$$P_{fi} = |c_f(T)|^2 \qquad\qquad (t \geq T)$$

of observing the system in state $|f\rangle$. We call this, the transition probability from state $|i\rangle$ to state $|f\rangle$. Notice that is is constant in time. The interaction potential shakes things up, mixing up the energy eigenstates, but once it turns off, things stop moving and stay in place.

Let's further assume that the interaction potenial V is small compared to the time-independent Hamiltonian $H^{(0)}$.

This scenario lends itself to perturbation theory, where the new interaction $\delta H(t)$ is refered to as the perturbration.

In the interaction picture, the well-understood Hamiltonian $H^{(0)}$ is generally large compared to the time-dependent additional interaction $\delta H(t)$. Starting from the state with no additional interaction (the unperturbed state) we calculate the effect of the perturbation as a series of corrections. Because our perturbation is ftime-dependent, we will need time-dependent pertubration theory. This topic is covered in PHY 115B, so we will not go into the topic too deeply. Fortunately, we can obtain the first-order results in a simple, intuitive manner. In the RHS of Equation 1, we make the approximation:

$$c_n(t) \sim c_n(0) = \delta_{ni}$$

to obtain a first-order estimate:

$$\dot{c_m}(t) = -\frac{i}{\hbar}\exp\left(i\frac{(E_m - E_i)t}{\hbar}\right)\langle m|\delta H(t)|n\rangle \tag{2}$$

For any final state $|f\rangle$, we can can estimate the corresponding coeffient for $t \geq T$ by an integral:

$$c_f(T) = -\frac{i}{\hbar}\,V_{fi}\int_o^T \exp\left(i\omega_{fi}t\right)\,dt$$

where we define the interaction potential matrix element:

$$V_{mn} = \langle m|V|n\rangle \tag{3}$$

and the angular frequency:

$$\omega_{mn} = \frac{E_m - E_n}{\hbar} \tag{4}$$

There is one subtlety, however. This prescription is only valid for $f \neq i$. At first order, we have

$$c_i(t) = 1.$$

In principle, $c_i(t)$ must diminish to account for the other coefficients becoming non-zero following Equation 2 as probablity goes as the square of the coefficients, this is a second order effect, zero at first order. Noting that:

$$\int_0^T e^i kx = e^{ikx}\frac{sin(kx/2)}{k/2}$$

we determine the first-order transition probability (for $f \neq i$) as:

$$P_{fi} = |c_f(T)|^2 = \frac{|V_{fi}|^2}{\hbar^2}\left(\frac{\sin(\omega_{fi}T/2)}{\omega_{fi}/2}\right)^2 \tag{5}$$

Notice that T appears in the numerator, but not the denominator. In the next section, we'll be looking closely at the factor:

$$F_T(\omega_{fi}) = \left(\frac{\sin(\omega_{fi}T/2)}{\omega_{fi}/2}\right)^2 \tag{6}$$

which we shall see saves the day for us, twice.

A Time-Dependent Perturbation Theory

Here we consider the simple time-dependence from Fig. 1. The new interaction is a time-independent interaction potential V which turns on a $t = 0$ and then off at $t = T$. When $t < 0$ the potential is off, and the total Hamiltonian is $H = H^{(0)}$. At $t = 0$, the potential turns on and $H = H^{(0)} + V$, and remains constant until $t = T$. For $t > T$, the potential is off, and $H = H^0$ again.

We assume that V is small, but specifically how small is TBD. We introduce a dimensionless time-independent factor λ as a bookkeeping device. When $\lambda = 1$, the perturbation is at full strength (but V is still small). When $\lambda = 0$ the perturbation is turned off. When we are finished with our bookkeeping[2] we will set $\lambda = 1$.

For $0 \leq t \leq T$ (that is, when the potential is on) we have:

$$\langle m|\delta H(t)|n\rangle = \lambda \langle m|V|n\rangle = \lambda V_{mn}$$

where we have defined:

$$V_{mn} \equiv \langle m|V|n\rangle$$

also defining:

$$\omega_{mn} \equiv \frac{E_m - E_n}{\hbar}$$

[2]If you prefer, you can take λ to be always small, by pulling out a small dimensionless parameter from V, then the TBD constraints will be on λV. I prefer to set λ to one once we are done using it for bookkeeping.

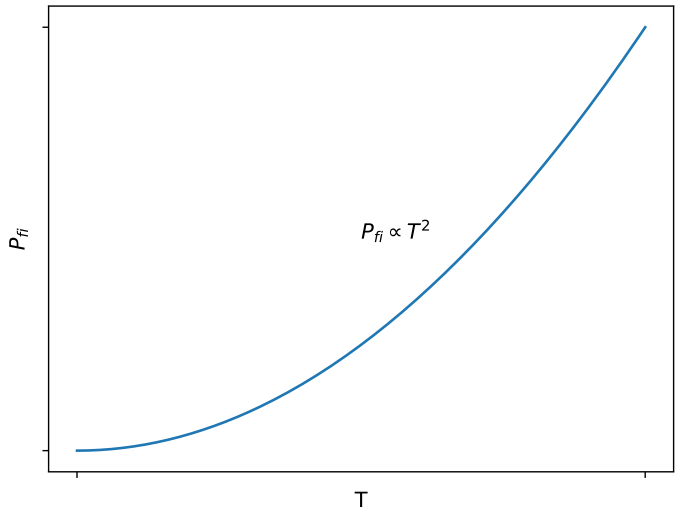

Figure 2: For $E_f = E_i$ the perturbation will fail (probality too large) for large enough T.

Equation 1 becomes:

$$\dot{c}_m(t) = -\frac{i\lambda}{\hbar} \sum_n c_n(t)\, e^{i\omega_{mn}t}\, V_{mn}$$

Notice that λ only appears on the RHS, which will be essential in what follows. Also, notice that for $\lambda = 0$ (perturbation off) we have

$$\dot{c}_m(t) = 0 \implies c_m(t) = \text{const}$$

as expected. Writing $c_m(t)$ as a power series in λ, we have:

$$c_m(t) \; = \; c_m^{(0)}(t) \; + \; \lambda\, c_m^{(1)}(t) \; + \; \lambda^2\, c_m^{(2)}(t) \; + \; \ldots$$

and thus:

$$\begin{aligned} \dot{c}_m{}^{(0)}(t) \; + \; \lambda\, \dot{c}_m{}^{(1)}(t) \; + \; \lambda^2\, \dot{c}_m{}^{(2)}(t) \quad &+ \quad \ldots \\ &= \quad -\frac{i}{\hbar} \sum_n \left\{ \lambda\, c_m^{(0)}(t) \; + \; \lambda^2\, c_m^{(1)}(t) \; + \ldots \right\} \, e^{i\omega_{mn}t}\, V_{mn} \end{aligned}$$

Matching terms of equal order in λ

$$\begin{aligned} \lambda = 0 \qquad &\to \; \dot{c}_m{}^{(0)}(t) &&= 0 \\ \lambda = 1 \qquad &\to \; \dot{c}_m{}^{(1)}(t) &&= -\frac{i}{\hbar} \sum_n c_m^{(0)}(t)\, e^{i\omega_{mn}t}\, V_{mn} \\ \lambda = 2 \qquad &\to \; \dot{c}_m{}^{(2)}(t) &&= -\frac{i}{\hbar} \sum_n c_m^{(1)}(t)\, e^{i\omega_{mn}t}\, V_{mn} \\ \lambda = N+1 \; &\to \; \dot{c}_m{}^{(N+1)}(t) &&= -\frac{i}{\hbar} \sum_n c_m^{(N)}(t)\, e^{i\omega_{mn}t}\, V_{mn} \end{aligned}$$

The first equation shows that:

$$c_m^{(0)}(t) = \text{const}$$

To start things off, note that with $\lambda = 0$ (the perturbation turned off)

$$c_m(t) = c_m^{(0)}(t) = \text{const}$$

which identifies $c_m^{(0)}(t)$ as the coefficients of the unperturbed system, which have no time dependence. If we assume that the initial state is $|i\rangle$, then:

$$c_m^{(0)} = \delta_{mi}$$

This has an immediate implication for our pertubrations. If we assume $c_i^{(1)}(t) = \epsilon$ for some small value ϵ then:

$$c_i(t) \approx c_i^{(0)}(t) + c_i^{(1)}(t) = 1 + \epsilon \approx 1$$

but for $m \neq i$, if we assume $c_m^{(1)}(t) = \epsilon$ for some small value ϵ then:

$$c_m(t) \approx c_m^{0}(t) + c_m^{(1)}(t) = 0 + \epsilon = \epsilon$$

So we will only consider higher order perturbations for $m \neq i$. In this case, the first order coefficient is:

$$\begin{aligned}
\dot{c_m}^{(1)}(t) &= -\frac{i}{\hbar} \sum_n c_n^{(0)}(t)\, e^{i\omega_{mn}t}\, V_{mn} \\
&= -\frac{i}{\hbar} \sum_n \delta_{ni}\, e^{i\omega_{mn}t}\, V_{mn} \\
&= -\frac{i}{\hbar}\, e^{i\omega_{mi}t}\, V_{mi}
\end{aligned}$$

and integrating over the period when the constant potential V is on:

$$\begin{aligned}
{c_m}^{(1)}(T) &= -\frac{i}{\hbar} \int_0^T dt\; e^{i\omega_{mi}t}\, V_{mi} \\
&= -\frac{i}{\hbar} V_{mi} \int_0^T dt\; e^{i\omega_{mi}t} \\
&= -\frac{1}{\hbar\omega_{mi}} V_{mi}\, \left(e^{i\omega_{mi}T} - 1\right) \\
&= -\frac{2i}{\hbar\omega_{mi}} V_{mi}\, \exp\left(i\frac{\omega_{mi}T}{2}\right) \sin\left(\frac{\omega_{mi}T}{2}\right)
\end{aligned}$$

The probability that the state $|i\rangle$ has transitioned into state $|f\rangle$ after the perturbation has been left on for time T is, to first order:

$$P^{(1)}_{\mathrm{f}\leftarrow\mathrm{i}} = |c_m^{(1)}(T)|^2 = \frac{|V_{fi}|^2}{\hbar^2}\; \frac{4\,\sin^2\left(\omega_{fi}T/2\right)}{\omega_{fi}^2} \tag{7}$$

As long as this probability is small, it should be a good approximation, and this is always possible for small enough V.

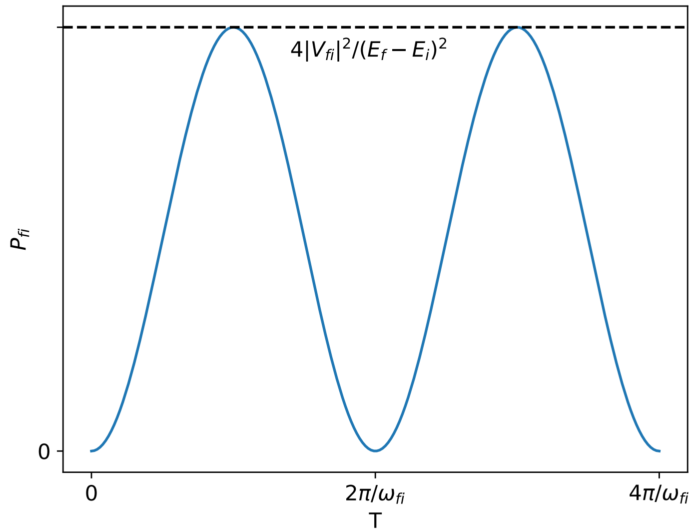

Figure 3: For $E_f \neq E_i$, the probability oscillates with an amplitude that goes as $1/(E_f - E_i)^2$

In the case that $E_f = E_i$ then we have $\omega_{fi} \to 0$, and:

$$P^{(1)}_{\mathrm{f}\leftarrow\mathrm{i}} = \frac{|V_{fi}|^2 T^2}{4\hbar^2}\mathrm{sinc}^2(\omega_{fi}T/2) \to \frac{|V_{fi}|^2 T^2}{4\hbar^2}$$

since

$$\mathrm{sinc}(x) \equiv \frac{\sin(x)}{x} \to 1$$

when $x \to 0$. As shown in Fig. 2, the probability grows quadratically with T, and so the perturbation will fail for larege enough T. So there is a limit to how large we can take T, but it can still get very large for small enough V. In the case that $E_f \neq E_i$, then we have the oscillatory behavior shown in Fig. 3. The oscillations have an amplitude:

$$\frac{4|V_{fi}|^2}{(E_f - E_i)^2}$$

which suppresses the probability of transitions that do not conserve energy. Evidently, turning on and off a constant potential does not change the energy much.

B Density of Final States

We cannot count continuum states, so we impose a fiction that a free particle is in fact constrained to a box with sides of length L and so:

$$V = L^3$$

We hope that our final results will not depend on this fictitious volume V. With this in mind, we consider the normalization of the energy eigenstates of the free particle (plane waves) as:

$$\psi(x) = Ne^{ik_x x}\, e^{ik_y y}\, e^{ik_z z}$$

If we choose:

$$N = \sqrt{\frac{1}{V}}$$

then:

$$\int_V |\psi(x)|^2 d^3x = 1$$

Assuming periodic boundary conditions:

$$\begin{aligned} k_x L &= 2\pi n_x \\ k_y L &= 2\pi n_y \\ k_z L &= 2\pi n_z \end{aligned}$$

so that

$$dn = dn_x dn_y dn_z = \left(\frac{L}{2\pi}\right)^3 dk_x dk_y dk_z$$

and we have:

$$dn = \left(\frac{L}{2\pi}\right)^3 d^3k \tag{8}$$

It is convenient to express Equation 8 as a differential in energy. As our box size cannot possibly be Lorentz invariant, we'll use the non-relativistic energy of the free particle (for now):

$$E = \frac{(\hbar k)^2}{2m}$$

so we have:

$$\begin{aligned} dk &= \frac{m}{\hbar^2 k} \\ d^3k &= k^2 dk d\Omega = \frac{mk}{\hbar^2} d\Omega dE \\ dn &= \left(\frac{L}{2\pi}\right)^3 \left(\frac{m}{\hbar^2 k}\right) d\Omega\, dE \end{aligned}$$

where we have identified the (classical) density of states:

$$\rho(E) \equiv \left|\frac{dn}{dE}\right| = \left(\frac{L}{2\pi}\right)^3 \left(\frac{m}{\hbar^2 k}\right) d\Omega \tag{9}$$

with:

$$dn = \rho(E)\, dE \tag{10}$$

C Fermi's Golden Rule

We now want to consider not just a transition from one discrete state to another discrete state, but instead, a transition from one discre state (the initial state) into a continuum state. We know that for the constant potential perturbation, final states with energy far from E_i will not contribute. So instead of counting discrete states will integrate over energy within ΔE of the initial state energy:

$$\sum_n \to \int dn = \int_{E_i - \Delta E}^{E_i + \Delta E} \rho(E) dE \equiv \int_{\Delta E} \rho(E) dE$$

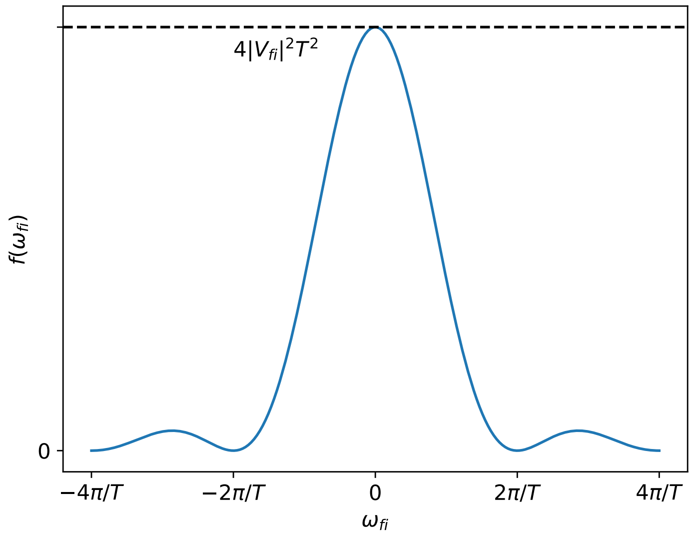

Figure 4: For large enough T, $f(\omega_{fi})$ becomes a delta function and the area under the curve is $\propto T$

where $\rho(E)$ is the density of states, for which we worked out an example previously.

So the total probability of a transition into one of these final states is:

$$\sum_f P^{(1)}_{f\leftarrow i}(T) \to \frac{4}{\hbar^2}\int |V_{\text{fi}}|^2\,\rho(E_f)\frac{\sin^2(\omega_{fi}T/2)}{\omega_{fi}^2}dE_f$$

The function

$$f(\omega_{fi}) \equiv \frac{sin^2(\omega_{fi}T/2)}{\omega_{fi}^2}$$

is plotted in Fig 4. Notice that it's integral will be approximately:

$$4|V_{fi}^2T^2 \times 4\pi/T \propto T$$

And also that it approaches a delta function for large enough T.

Therefore, we can take $|V_{fi}|^2$ and $\rho(E_f)$ outside of the integral, by evaluating them at $E_f = E_i$, and we can extend the integral out to infinity:

$$\sum_f P^{(1)}_{f\leftarrow i}(T) \to \frac{4}{\hbar^2}|V_{\text{fi}}|^2\,\rho(E_i)\int_{-\infty}^{\infty}\frac{\sin^2(\omega_{fi}T/2)}{\omega_{fi}^2}dE_f$$

The integral can now be evaluated, leading to:

$$\sum_f P^{(1)}_{f\leftarrow i}(T) \to \frac{2\pi}{\hbar}|V_{fi}|^2\rho(E_i)T$$

This allows us to calculate the rate of transitions from a discrete state $|i\rangle$ into a continuum state $|f\rangle$ as:

$$\Gamma^{(0)}_{fi} = \frac{\sum_f P^{(1)}_{f\leftarrow i}(T)}{T} = \frac{2\pi}{\hbar}|V_{fi}|^2\rho(E_i)$$

This is Fermi's Golden Rule (to first order) the foundation of experimental particle of physics.

When we include higher order terms from TDPT, Fermi's Golden Rule becomes:

$$\Gamma_{fi} = \frac{2\pi}{\hbar} |T_{fi}|^2 \rho(E_i)$$

where

$$T_{fi} = \langle f|T|i\rangle = \langle f|V|i\rangle + \sum_{k\neq i} \frac{\langle f|V|k\rangle \langle k|V|i\rangle}{E_i - E_k} + \ldots$$

is called the transition matrix element.

Recall that:

$$\rho(E) \equiv \left|\frac{dn}{dE}\right|$$

Fermi's Golden rule takes the simple form it does because energy is conserved, and so we took $\rho(E)$ to be a constant outside of the integral by putting $E_f = E_i$. But we can put it back into the integral with delta function to impose energy conservation:

$$\rho(E_i) = \left|\frac{dn}{dE}\right|_{E_i} = \int \left|\frac{dn}{dE_f}\right| \delta(E_i - E_f) dE_f = \int \delta(E_i - E_f) dn$$

So we have an equivalent form of Fermi's Golden Rule:

$$\Gamma_{fi} = \frac{2\pi}{\hbar} \int |T_{fi}|^2 \delta(E_i - E_f) dn \tag{11}$$

This has a very nice interpretation. We integrate over the entire phase space (dn) of the final state, subject to energy conservation, and weight each region of phase space by the transition matrix. The phase space is just the kinematics (eventually relativistic kinematics) and the transition matrix contains all of the dynamics.

D Lorentz Invariant Phase Space

The density of final states that we calculated in Section B is not Lorentz Invariant. We'll first show that the four-momentum volume integral:

$$d^4p = dp^0 dp^1 dp^2 dp^3 = \frac{1}{c} dE dp_x dp_y dp_z$$

is Lorentz Invariant.

Consider a Lorentz transformation Λ between frames so that (in matrix representation) we have:

$$[p]_{\mathcal{B}'} = \Lambda [p]_{\mathcal{B}}$$

Then the Jacobian (see next section if this is unfamiliar) for the coordinate transformation is:

$$J \equiv \det \Lambda$$

and the four-momentum volume as calculated in the frame F' is:

$$(d^4p)' = |\det \Lambda|\, d^4p$$

However, the defining property of Lorentz transformations is:

$$\Lambda^T G \Lambda = G$$

so using properties of the determinant:

$$\begin{aligned}
\det\left(\Lambda^T G \Lambda\right) &= \det G \\
\det(\Lambda)\det(G)\det(\Lambda) &= \det G \\
(\det(\Lambda))^2 \det(G) &= \det G
\end{aligned}$$

from which it follows that:

$$\det \Lambda = \pm 1$$

Therefore we have:

$$(d^4p)' = |\det \Lambda|\, d^4p = d^4p$$

and the four-momentum volume is Lorentz Invariant.

But an integral over d^4p will included many different particle masses via:

$$m^2c^2 = p^\mu p_\mu = (p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2$$

and we want the phase space of a final state particle with a certain mass. So we consider a phase-space integral:

$$dPS = \delta(p^\mu p_\mu - m^2c^2)\; d^4p$$

But this still allows $E = p^0c$ to run over negative values, so we include:

$$dPS = \theta(p^0)\; \delta(p^\mu p_\mu - m^2c^2)\; d^4p$$

This means dPS is not invariant wrt time reversal, but it is still invariant with respect to the **restricted** Lorentz Group which does not include Parity (P) and Time-Reversal (T). Lastly, we'll put the factors of 2π, consistent with what we calculated in Section B:

$$dPS = \theta(p^0)(2\pi)\delta(p^\mu p_\mu - m^2c^2)\; \frac{d^4p}{(2\pi)^4}$$

This form of dPS makes is Lorentz Invariance manifest. However, there is another way to write it, by noting:

$$\delta(p^\mu p_\mu - m^2c^2) = \delta((p^0)^2 - |\vec{p}|^2 - m^2c^2) = \delta((p^0)^2 - (E/c)^2)$$

and using this to complete the integral over dp^0. Note the delta function identity (proven below) that:

$$\delta(x^2 - a^2) = \frac{1}{2|a|}\left(\delta(x-a) + \delta(x+a)\right)$$

In this case, we have $p^0 > 0$ so only the first δ function matters, and we have:

$$dPS = \frac{d^3p}{(2E/c)\;(2\pi)^3}$$

When working with integrals of phase space, we have to be somewhat careful, and recall that E is an integration variable that has been set (via the delta function) to the value $E = \sqrt{m^2c^4 + |\vec{p}|^2c^2}$ which depends on the other integration values!

This differs from the density of final states calculated earlier via only constants and one dynamical quantity $2E$.

E Lorentz Covariant Normalization of the wave functions

Suppose that a particle of mass m is contained in a container C. The rest frame of the container is frame F, in which the container has volume V_0. The probability of finding the particle in container C is one, so:

$$\int_C |\psi(x)|^2 d^3x = 1$$

The number of particles per unit volume is therefore:

$$\frac{1}{V_0}\int_C |\psi(x)|^2 d^3x = \frac{1}{V_0}$$

In frame F', moving with respect to the container, the volume of the container is contracted by:

$$V' = V_0/\gamma$$

But the probabililty of the particle being in the container is still one, so the number of particle per unit volume as caculated in frame F' is therefore:

$$\frac{1}{V'}\int_A |\psi(x)|^2 d^3x = \frac{1}{V'} = \frac{\gamma}{V_0}$$

Clearly, than, normalizing to the number of particle per unit volume is not Lorentz invariant.

We found in the previous section that our phase-space term was missing a factor of $2E$. Let's look at what happens if we normalize an energy eigenstate to $2E$ particles per unit volume. For illustration, let's assume that the volume V is large enough that the particle can be taken to be at rest in frame F (without violating the uncertainty principle). Then we will have, in frame F:

$$\frac{1}{V_0}\int_C |\psi(x)|^2 d^3x = 2E = 2mc^2$$

Meanwhile, in frame F', we must find:

$$\frac{\gamma}{V_0}\int_C |\psi(x)|^2 d^3x = 2\gamma mc^2 = 2E'$$

Thus, we will now adopt the convention of normalizing energy eigenstates to $2E$ particles per unit volume, which is a Lorentz invariant convention.

Just to be clear: this means that a wave function normalized to $2E$ particles per unit volume in frame F will also be normalized to $2E'$ particles in frame F', even though these will, generally, be different values.

F Lorentz Covariant Form of Fermi's Golden Rule

We derived the non-relativistic form of Fermi's Golden Rule as:

$$\Gamma_{fi} = \frac{2\pi}{\hbar}\int |T_{fi}|^2 \delta(E_i - E_f) dn$$

We'll now look at how to adapt Fermi's Golden Rule for Lorentz covariance and muliple initial state particles, which we will write as:

$$a + b + c + \ldots \;\rightarrow\; 1 + 2 + 3 + \ldots + n$$

So in: $T_{fi} = \langle f|T|i\rangle$ we now have:

$$|i\rangle = |\Phi_a\Phi_b\Phi_c\ldots\rangle$$

and

$$\langle f| = \langle\Phi_1\Phi_2\Phi_3\ldots\Phi_n|$$

where each initial state wave function (e.g. Φ_a) and each final state wave function (e.g. Φ_1) are normalized in the non-relativistic fashion: one particle per unit volume. We have therefore:

$$T_{fi} = \langle\Phi_1\Phi_2\Phi_3\ldots\Phi_n|T|\Phi_a\Phi_b\Phi_c\ldots\rangle$$

Now, instead of T_{fi}, let us instead consider a Lorentz invariant matrix element $\mathcal{M}_{fi}$ calculated using basis vectors noramlized in the Lorentz invariant fasion, e.g.:

$$\Psi_a = \sqrt{2E_a}\Phi_a$$

and so on. So:

$$\begin{aligned}
\sqrt{c^{n-1}\hbar^{3(m-1)}}\mathcal{M}_{fi} &\equiv \langle\Psi_1\Psi_2\Psi_3\ldots\Psi_n|T|\Psi_a\Psi_b\Psi_c\ldots\rangle \\
&= \sqrt{2E_1\;2E_2\;2E_3\;\ldots}\,\langle\Phi_1\Phi_2\Phi_3\ldots|T|\Phi_a\Phi_b\Phi_c\ldots\rangle\,\sqrt{2E_a\;2E_b\;2E_c\;\ldots} \\
&= \sqrt{2E_1\;2E_2\;2E_3\;\ldots}T_{fi}\sqrt{2E_a\;2E_b\;2E_c\;\ldots}
\end{aligned}$$

we have included a factor of $\sqrt{c^{n-1}}$ (for n final state particles) and $\sqrt{\hbar^{3(m-1)}}$ (for m initial state particles) to match the conventions for the Feynman Rules that which we will be applying to determine $\mathcal{M}$. And so:

$$|T_{fi}|^2 = \frac{|\mathcal{M}_{fi}|^2}{(2E_a\;2E_b\;2E_c\;\ldots)}\frac{c^n}{(2E_1\;2E_2\;2E_3\;\ldots 2E_n)}$$

Notice that the new normalization has led to exactly the factors of $1/2E$ needed to make each phase space factor Lorentz Invariant.

Thus, to construct a Lorentz Covariant form of Fermi's Golden Rule appropriate to a process with multiple intitial and final state particles, we make the following replacements

$$|T_{fi}|^2 \longrightarrow \frac{|\mathcal{M}_{fi}|^2}{c\;(2E_a\;2E_b\;2E_c\;\ldots)} \tag{12}$$

and

$$(2\pi)\;\delta(E_i - E_f)\;dn \longrightarrow (2\pi)^4\delta^4(p_a + p_b + \cdots - p_1 - p_2 - \ldots) \times \Pi_i\,\frac{c\;d^3p_i}{2\;E_i(2\pi)^3} \tag{13}$$

Remember that each:

$$E_i \equiv \sqrt{m_i^2c^4 + p_i^2c^2}$$

which is a function of the integration variable. These are not factors that can be simply pulled out of the integral!

The phase space for n final state particles is completely covered by integrating over the first $n-1$ particles, because it's momentum is completely defined by the first $n-1$ particles due to momentum conservation. However, we have opted to integrate over all n final state particles, and include a δ function imposing momentum conservation, which is equivalent.

G Math Tools

In the previous section, we used some math tools. First, we need to know how to calculate the Jacobian of a coordinate transformation. Suppose we are calculated an integral:

$$I = \int f(u, w)\, du\, dw$$

but suppose we have some other coordinates r and s with:

$$\begin{aligned} u &= u(r, s) \\ w &= w(r, s) \end{aligned}$$

then we can calculate our integral in terms of coordinates r and s via:

$$I = \int f(u(r, s), w(r, s)) \left| \frac{\partial(u, w)}{\partial(r, s} \right| dr\, ds$$

where

$$\frac{\partial(u, w)}{\partial(r, s)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial s} \\ \frac{\partial w}{\partial r} & \frac{\partial w}{\partial s} \end{vmatrix}$$

is the Jacobian: the determinant of the matrix of partial derivatives. We can summarize this as:

$$du\, dw = \left| \frac{\partial(u, w)}{\partial(r, s)} \right| dr\, ds$$

The result easily extends to more coordinates, e.g.:

$$du\, dv\, dw = \left| \frac{\partial(u, v, w)}{\partial(r, s, t)} \right| dr\, ds\, dt$$

where now the Jacobian will be the determinant of 3x3 matrix.

Second, we needed:

$$\delta(x^2 - a^2) = \frac{1}{2|a|} \left(\delta(x - a) + \delta(x + a)\right)$$

There's a trick for finding indentities like this, using the Heaviside step function:

$$\theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}$$

and noting that

$$\int_{-\infty}^{x} \delta(y) dy = \theta(x)$$

so we can consider the delta function to be the derivative of the Heaviside step function:

$$\delta(x) = \theta'(x)$$

To find our identity, we work out that:

$$\theta(x^2 - a^2) = \theta(x - |a|) + (1 - \theta(x + |a|))$$

Take the derivative of both sides:

$$\begin{aligned}\theta'(x^2-a^2)2x &= \theta'(x-|a|)-\theta'(x+|a|)\\ \theta'(x^2-a^2) &= \frac{\theta'(x-|a|)}{2x}-\frac{\theta'(x+|a|)}{2x}\end{aligned}$$

Now identify $\delta(x)$ with $\theta'(x)$ and simplifying:

$$\begin{aligned}\delta(x^2-a^2) &= \frac{\delta(x-|a|)}{2x}-\frac{\delta(x+|a|)}{2x}\\ &= \frac{\delta(x-|a|)}{2|a|}-\frac{\delta(x+|a|)}{-2|a|}\\ &= \frac{\delta(x-|a|)+\delta(x+|a|)}{2|a|}\\ &= \frac{\delta(x-a)+\delta(x+a)}{2|a|}\end{aligned}$$

H Fermi's Golden Rule for Two-Body Decays

For a decay process $a \to 1+2$, the Lorentz invariant form of FGR is:

$$\Gamma_{fi} = \frac{1}{\hbar\, c\, (2E_a)} \int |\mathcal{M}_{fi}|^2\, (2\pi)^4 \delta^4(p_a - p_1 - p_2) \frac{c\, d^3p_1}{2E_1(2\pi)^3} \frac{c\, d^3p_2}{2E_2(2\pi)^3}$$

where we must as always remember that $E_1 = \sqrt{p_1^2c^2 + m_i^2c^4}$. Notice that

$$\Gamma_{fi} \propto \frac{1}{2E_a} \propto \frac{1}{\gamma}$$

where γ is the expected time dilation factor from special relativity.

I Fermi's Golden Rule for Two-Body Scattering